

PROJECT SUTRA — NATURAL PITCH SCRIPT

PLAIN-ENGLISH SPOKEN DELIVERY • HIGH-TECH VERIFIED ON SLIDES (NIKHIL & HARIKA)

PRESENTERS: NIKHIL • HARIKA
TOTAL TIME: ~2 MIN 15 SEC

💡 **CORE RULE:** Speak the *human impact & physical reality*. Never read raw technical acronyms aloud. Let the slide prove the engineering rigor (Deep JSCC, 50Hz MicroXRCE, WGS84 DEM) while you explain *what it actually does*.

SLIDE 01 Introduction & Core Value

00:00 - 00:30 (30s)

SPEAKER: NIKHIL (TECH LEAD) [STAGE: CENTER-LEFT • CONFIDENT & CALM]

🔥 **HOOK:** "When a disaster strikes, the first 72 hours decide who lives. But the second drones enter thick forests or mountain ravines, they lose signal and crash."

"Respected jury, we are **Team OFFGRID**. I am Nikhil, and this is Harika. We built **Project SUTRA** — an autonomous drone swarm that finds survivors in disaster zones with **zero GPS and zero internet**."

"Instead of single manual drones, SUTRA deploys an intelligent swarm backed by 3 breakthroughs:"

PILLAR 1 Autonomous Swarm Flight: Drones dodge obstacles and coordinate together 50 times a second without ground towers. [Slide: 50Hz GNC]

PILLAR 2 AI-Powered Comms: Live video stays connected through mountain walls without blacking out. [Slide: -5dB JSCC]

PILLAR 3 3D Elevation Vision: Locks onto survivors on steep slopes with centimeter accuracy. [Slide: 3.59cm DEM]

HANDOFF TO HARIKA → "Harika will explain why current search and rescue missions fail in the field."

SLIDE 02 The Problem: 4 Field Failure Voids

00:30 - 01:15 (45s)

SPEAKER: HARIKA (OPERATIONS & AUDIT LEAD) [STAGE: STEP FORWARD • CLEAR & FACTUAL]

⚠️ **HOOK:** "Thank you, Nikhil. When we audited real NDRF disaster reports, we found 4 fatal breakdowns that cause current rescue drones to fail:"

VOID 1 Tree Canopies Block GPS: Drones lose satellite lock, drift blindly into branches, and crash. (*In Wayanad, 70% of canopy sorties failed*).

VOID 2 Mountain Ridges Cut Video: The moment a drone dips behind a ridge, standard video glitches and cuts to total black.

VOID 3 Flat-Map Errors on Slopes: Standard drone cameras assume the earth is flat. On steep hillsides, coordinates miss by **15 to 30 meters** — routing teams to empty ravines.

VOID 4 Human Pilot Overload: Flying 5 drones takes **15 to 25 crew members**, taking hours to set up and costing ₹12.5 Lakhs per sortie.

HANDOFF TO NIKHIL → "Nikhil, show the jury how SUTRA fixes this."

NIKHIL: AUTONOMOUS FLIGHT & COMMS [20s • CONFIDENT ARCHITECT]

"SUTRA replaces these breakdowns with 4 engineering moats:"

- MOAT 1**Self-Flying Swarms:** Drones calculate collision-free flight paths onboard at 50Hz — zero ground towers, zero GPS reliance. [Slide: SUTRA-FSD & 3D ORCA]
- MOAT 2**AI-Guided Neural Video:** Even in deep ravines where standard signals die, our neural link transmits clear video through solid mountain terrain. [Slide: Deep JSCC | -5dB]

HARIKA: 3D VISION, COMMAND & BENCHMARKS [40s • CLOSING PUNCH & ROI]

- MOAT 3**3D Mountain Vision:** Onboard AI maps terrain height in real time, locating survivors within **3.5 centimeters** in under 15 milliseconds. [Slide: 3D DEM Jetson Orin]
- MOAT 4**One-Person Command:** A single tactical tablet lets **1 operator command the entire swarm** for just **₹95,000 per sortie**. [Slide: Pegasus 3D GCS]

"The bottom line: SUTRA scans a full square mile in 10 minutes — 4 times faster than human rescue crews, saving lives before time runs out."

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min (4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000